USL Program — Paper Alignment Table

Corpus classification by directness of alignment with the Universal Structural Law · all 18 papers

PAPER	ROLE IN PROGRAM	USL ALIGNMENT	ACTION
— DIRECT / CORE —			
The Universal Structural Law v6	Central synthesis: states the ladder inequality, corpus result, adversarial tests, percolative scope boundary, and full UNNS interpretation layer stack.	Direct · core	**Keep** Enforce strict Layer 1/2/3 separation throughout; all other papers are positioned relative to this.
— HIGH / FOUNDATIONAL BRIDGE —			
Structural Lawhood as Interior Admissibility Phase geometry of operator families across domains	Introduces the rigidity modulus R and a sharp phase boundary separating stable and degeneracy-admissible regimes. Proves the matching-number theorem and gives seismic empirical instantiation. Cross-domain universality claim.	High · foundational bridge	**Keep · revise wording** Degeneracy-admissible must not be mistaken for stable admissibility. $R>1 \to$ interior; $R<1 \to$ structural change is not prevented.
Perturbation-Admissible Structural Stability in the UNNS Substrate Admissibility geometry, phase boundaries, quotient descent	Formalises perturbation-admissible structural stability via two rigidity moduli (R_{mag}, R_{geo}). Proves $R>(1,1)$ sufficient for full descent (Order Rigidity + Directional Stability) and derives a Phase-Boundary Theorem. Instantiated on seismic chamber suite LXV.	High · foundational bridge	**Keep** Closest operator-theoretic precursor to USL; core of the admissibility spine that USL concretises in ordered ladders.
Operator–Manifold Admissibility Geometry Theoretical framework paper (companion to Axis VI)	General admissibility-manifold language for operator families and structural sectors. Defines the conceptual container in which USL sits.	High · framework bridge	**Keep** Provides the overarching geometric vocabulary; cite as container, not as direct derivation of the ladder inequality.
On the Structural Position of the Structural Lawhood Framework Within the UNNS substrate program	Clarifies the structural relationship between the LI–LV arc, the Structural Lawhood manuscript, and the Axis VI empirical program. Shows the three components share a common invariant: stratified operator-manifold phase geometry.	Med-high · interpretive bridge	**Keep · scope-limit** Good for conceptual placement and bridging. Must not advance stronger empirical claims than USL itself supports.

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Admissibility Constraints Across Physical and Biological Systems Empirical extension of USL — QT45 ribozyme via STRUC-I v1.0.4 and STRUC-BIO-I v0.1	Extends USL validation to evolutionary biology using two independent instruments applied to the QT45 self-replicating RNA polymerase ribozyme. Zero violations across 240 STRUC-I evaluations of 6 fitness ladders and 3 STRUC-BIO-I configurations. Substitution ladders cluster at GOE-nuclear tier (rho_bar 0.081-0.277). Deletion ladders yield first Boundary-Stabilized physical biological result (rho_bar=0.819), highest structural pressure for any physical biological system in the corpus. Clean QT45 epistasis network is sub-GOE (rho=0.032, z=+4.22). Cross-instrument consistency (STRUC-I vs STRUC-BIO-I) is the strongest available evidence for domain independence of the admissibility constraint. Total corpus extended to 3,313 evaluations with zero clean violations.	High · empirical extension	Keep · scope carefully First biological domain validation; significantly strengthens USL cross-domain scope claim. Limitations: single system only; percolative realizability unproved for biological landscapes; fitness noise uncharacterised.
— MEDIUM / EMPIRICAL SUPPORT & DOMAIN INSTANTIATION —			
Operator–Manifold Admissibility Geometry Cross-domain empirical certification in seismology and cosmology (Axis VI)	Tests the primary admissibility falsifier exhaustively across seismic displacement fields (3 earthquakes, 5 smoothing windows) and CMB angular power spectra (3 polarisation channels). Falsifier never triggered. Dense boundary stratification in cosmology; event-dependent stratification in seismology.	Medium · empirical support	Keep · frame carefully Domain-specific empirical certification of admissibility geometry, not a direct proof of the USL ladder inequality.
UNNS Chambers Certification Report Chambers LVI · LVII · LVIII · LVIX · LX — Protocol v1.0.0	Statistical certification of five convex sign-preservation chambers under fully locked protocol. Establishes consistent operator family hierarchy (V2×V3 compounding, V2×V4 near-isometric, V6×V7 intermediate). Supports the calibration-floor obstruction result.	Med-high · empirical support	Keep Provides the chamber-level statistical backbone for the sign-preservation cluster. Treat as supporting evidence, not as a standalone theorem paper.
Chamber LV Certification Report Tier-2 selective transfer — structural assessment	Tier-2 lifting result: normalization null (encoding hypothesis falsified), slope stratification confirmed, channel orthogonality preserved (R^2=0.0000 across all families/tiers). Feeds directly into the Structural Completeness unification paper.	Med-high · empirical support	Keep Concrete falsifier output that grounds the LI–LV arc; best cited in conjunction with the Structural Completeness paper.
Directional Rigidity Under Multipole Truncation CMB operator chart extension: axis geometry at low ■	Defines the admissible truncation operator chart on CMB harmonic coefficients. Tracks the quadrupole axis under multipole cutoff. Demonstrates axis stability as a structural invariant across the operator chart.	Medium · domain instantiation	Keep Supports cross-domain recurrence of the same structural geometry; good evidence that rigidity logic is not ladder-specific.
Directional Rigidity of Planetary Gravity Fields Under harmonic extension — Earth, Mars, Moon	Tracks the dominant orientation axis u(L) for Earth (EIGEN-6C4), Mars (JGM85F01), and Moon (AIUB-GRL350A) across full harmonic expansions. All three bodies: axis stabilises at L=2, zero total path. Synthetic random field accumulates 25.84° vs. 0°. Spectral gaps 3.4–7.9× larger than synthetic control.	Medium · domain instantiation	Keep Strong empirical evidence that physically realised fields occupy deep interior regions of the admissibility manifold; extends rigidity logic beyond seismology.
— INDIRECT / WORLDVIEW FOUNDATION —			

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The UNNS Observability–Admissibility Duality Theorem	Proves admissibility operators and observability projections form a duality: admissibility may reveal latent structure while observability may erase it exactly. Intended to resolve apparent contradictions between null experimental results and underlying dynamics.	Indirect · worldview foundation	**Keep · separate** Supports the substrate worldview and Layer 3 interpretation. Does not derive the ladder inequality; must be clearly decoupled from USL's proof structure.
Factorization Inevitability in Recursive DAG Admissibility	Proves admissibility operators over recursive substrate dynamics necessarily factorize into four independent structural constraint channels (Tier-1, Chambers XLVII–L). Correspondence theorem: any genuinely new gate either refines existing DAG-level invariant classes or extends the substrate model.	Indirect · worldview foundation	**Keep** Deep substrate-side architecture. Good as worldview foundation; do not cite as direct USL lineage.
Structural Completeness, Robustness, and Hierarchical Non-Isometry Unified theory of admissibility geometry under operator lift — Chambers LI–LV	Consolidates LI–LV arc into a unified theory. Proves Projection Non-Isometry Theorem (admissibility partitions preserved under lift; geometry not uniformly metric-preserving). Categorical formulation via Lawvere-metric enrichment. Introduces Retention Slope Index (R5-T) as computable stratification signature.	Indirect · worldview foundation	**Keep** Foundational admissibility and persistence machinery. Do not cite as a direct ancestor of the ladder theorem.
— INDIRECT / ANALOGICAL & SUPPORT MATERIAL —			
Curvature Sign Preservation Under Admissible Recursion Chambers LXI–LXIV: empirical elimination to formal proof scaffold	Four chambers probe every mechanistic hypothesis for certified negative curvature across 7,400+ cell-regime records: zero cert_neg cells observed. Translates multi-axis elimination into a six-ingredient formal proof scaffold. Strengthened admissibility axiom elevates xγ-vertex positivity from open lemma to axiom.	Indirect · analogical support	**Keep · with caution** Analogical reinforcement of 'admissibility first' via sign-preservation channel. Not a direct derivation of USL; empirical phase is complete but formal proof scaffold has open obligations.
Admissibility, Depth Submultiplicativity, and Universal Sign Preservation	Identifies minimal structural conditions under which recursive operator dynamics cannot certify persistent negative curvature. Central mechanism: depth submultiplicativity of a dispersion functional plus a positive margin condition. Universal Sign Preservation Theorem stated independently of any specific numerical constant.	Indirect · analogical support	**Keep · with caution** Provides the theoretical substrate for the sign-preservation cluster. Analogical rather than direct; empirical constants treated separately as certification data.
Convex Sign Preservation and the Calibration-Floor Obstruction Recursive curvature dynamics — Chambers LVI–LX	Analyses convex kernel recombination and gain scaling across five chambers. No certified negative-curvature region detected despite strongly negative point estimates (bmin ≈ -0.30). Identifies calibration-floor obstruction: negative-curvature region confined to high-variance boundary stratum where non-degeneracy gate cannot be cleared. Formalises Convex Sign Preservation Conjecture with partial theorem-level statement.	Indirect · analogical support	**Keep · with caution** Provides geometric-statistical rigidity evidence analogical to USL's admissibility logic. Conjecture is not yet a full theorem; scope must be stated clearly.

PAPER	ROLE IN PROGRAM	USL ALIGNMENT	ACTION
struc_i_v1_0_4_corpus_analysis.html Corpus dashboard / analysis artifact	Dashboard summarising chamber outputs and key metrics across domains. Narrative layer over the formal results.	**Direct as evidence artifact**	**Revise** Best treated as an evidence / report layer companion to USL. Narrative language can drift into stronger claims than the formal paper supports; scope carefully before public release.